

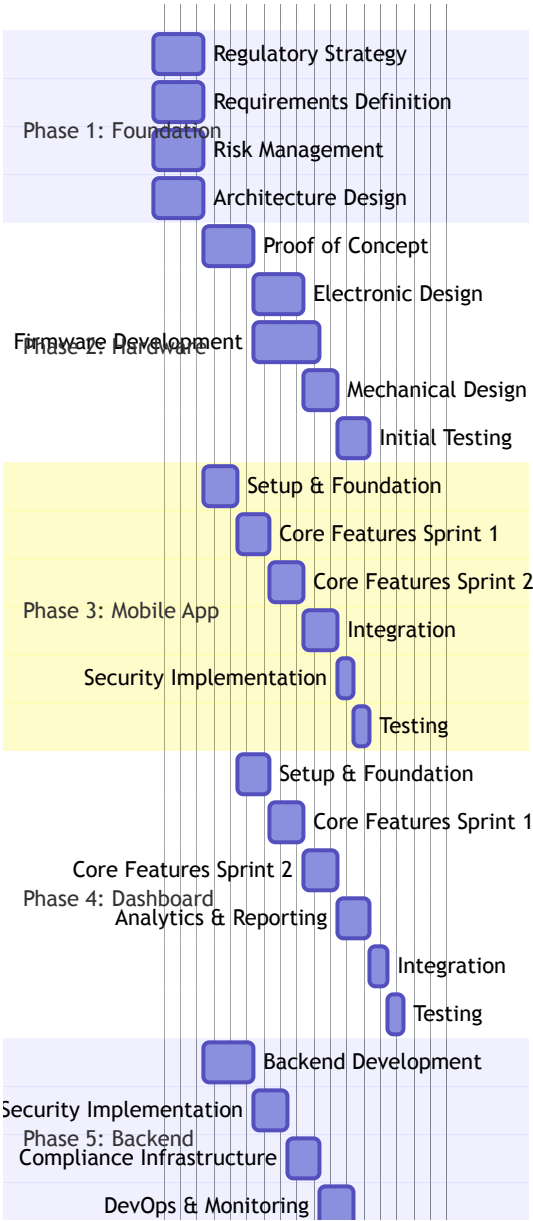
Medical Device MVP Development

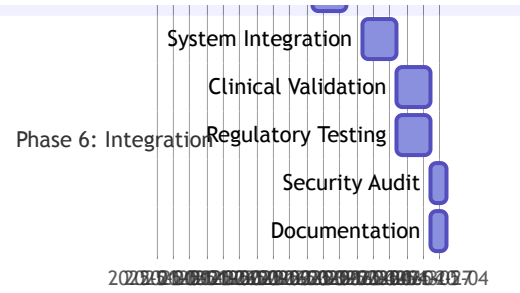
Workflow Diagrams - Ultrasound Probe + Mobile App + Dashboard System

1. Overview Timeline (Gantt Chart)

This Gantt chart shows the complete timeline for all 6 development phases, illustrating parallel workstreams and dependencies. The project spans approximately 16 weeks from foundation to integration.

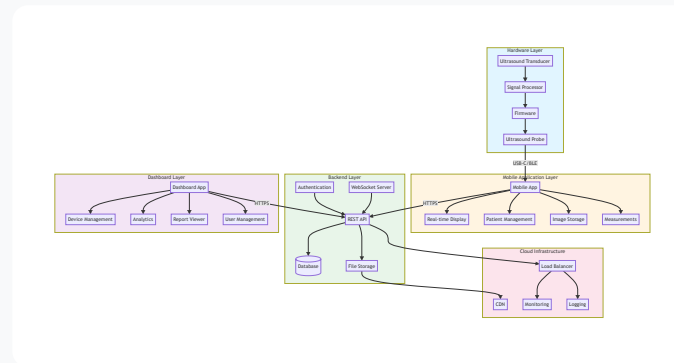
Medical Device MVP Development Timeline





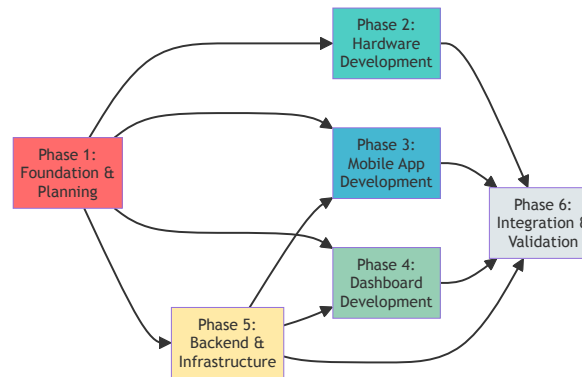
2. System Architecture Flow

High-level system architecture showing the four main layers (Hardware, Mobile, Backend, Dashboard) and cloud infrastructure. Arrows indicate data flow and communication protocols between components.



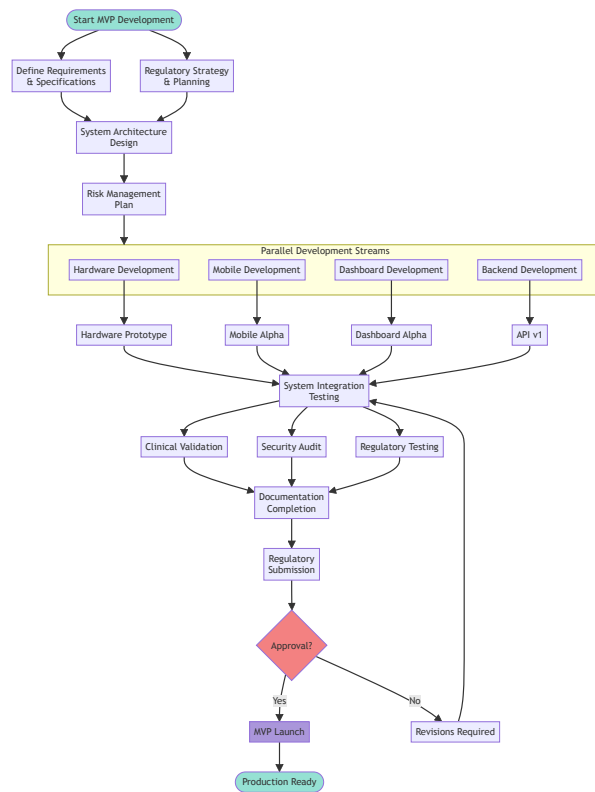
3. Phase Dependencies

Visual representation of dependencies between development phases. Phase 1 (Foundation) is a prerequisite for all other phases. Phase 5 (Backend) must be established before mobile and dashboard development can proceed to integration.



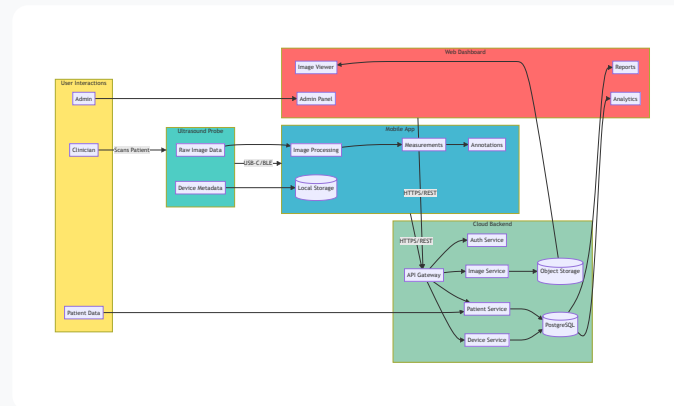
4. Development Workflow Process

End-to-end development workflow from project initiation through regulatory approval and market launch. Shows parallel development streams and the iterative nature of regulatory submission.



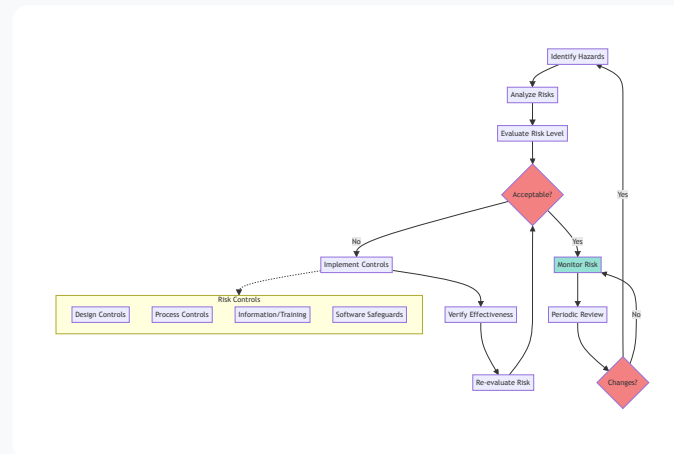
5. Data Flow Diagram

Detailed data flow showing how patient data, images, and device metadata move through the system from clinician interactions to cloud storage and dashboard viewing.



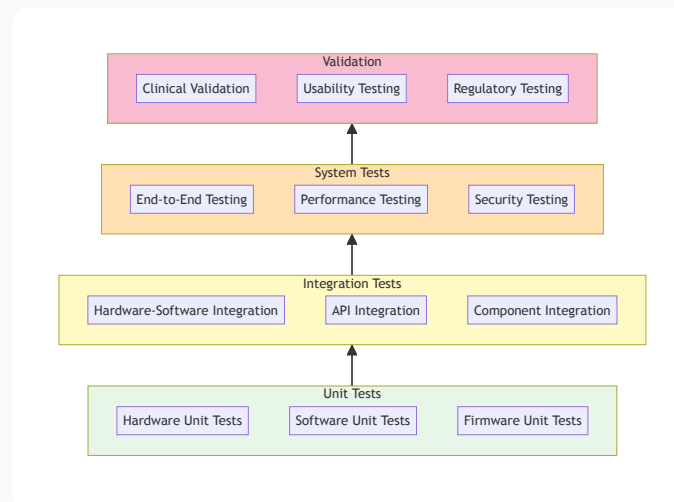
6. Risk Management Flow

ISO 14971 compliant risk management process showing continuous identification, analysis, evaluation, control implementation, and monitoring of risks throughout the product lifecycle.



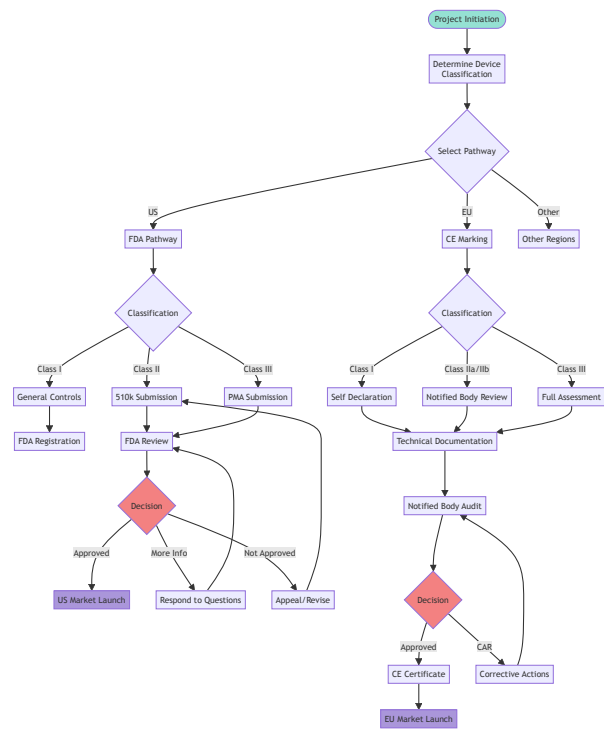
7. Testing & Validation Pyramid

Hierarchical testing approach from unit tests at the base through integration and system testing to clinical validation at the apex. Each level builds upon the previous to ensure comprehensive quality assurance.



8. Regulatory Pathway

Decision tree for regulatory clearance showing FDA pathways (510(k), PMA) and CE marking routes based on device classification. Includes review cycles and approval decision points.



9. Sprint Structure (Agile)

Two-week Agile sprint cycle showing parallel workstreams for design, development, testing, and documentation. Includes sprint planning, review, and retrospective ceremonies.

2-Week Sprint Cycle

